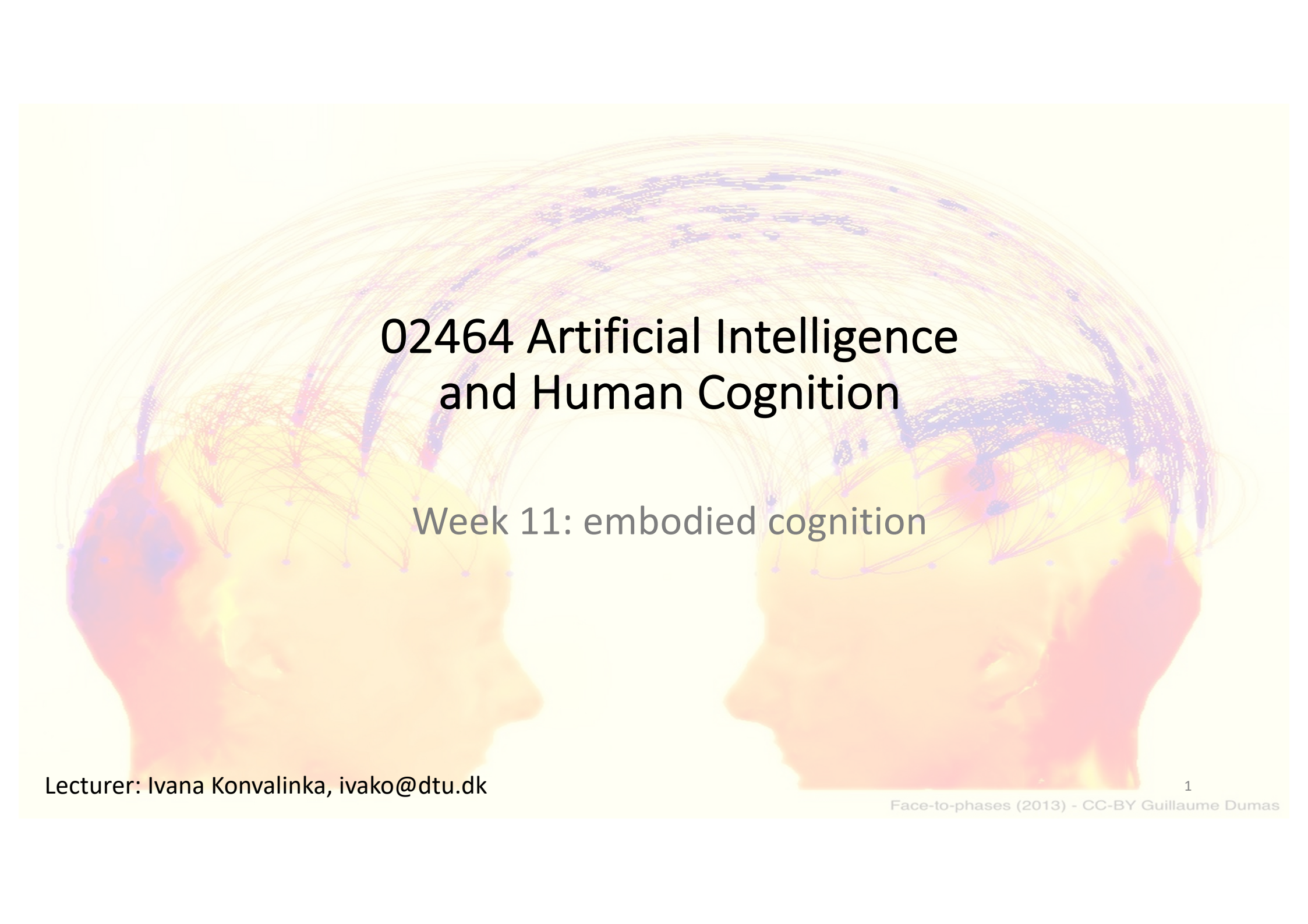




02464 Artificial Intelligence and Human Cognition

Week 11: embodied cognition

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Examples of “theoretical” exam questions

Motor control

- Imagine you are navigating through a dimly lit room toward a door on the opposite side. Your vision is limited, making it challenging to discern exact details or distances.
 1. Describe, using Bayesian principles, how your brain might integrate sensory information and other relevant factors to guide your movement toward the door.
 2. How might your brain’s approach change if the lighting were significantly brighter?
- Answer:
 1. The brain will optimally combine priors and likelihood. In the dimly lit room, it will increase the weight of prior knowledge (past experiences of room layouts or typical walking distances) because the current sensory input (the likelihood) is unreliable.
 2. For brighter lighting, the sensory input is more reliable, hence the brain may rely more on sensory information, adjusting the weight of priors accordingly.

Examples of “theoretical” exam questions (from last year’s exam)

Motor control

- You are playing tennis with a friend. You know which way you want the tennis ball to travel when you return it to your opponent, but what is the right way to hit the ball? An inverse model of motor control is thought to aid in this process. Describe how your brain trains your inverse model to hit the tennis ball back to your opponent.

- Answer:

When you want to hit the tennis ball back to your opponent, you specify a desired state, the ball’s intended outgoing direction. Your inverse model must generate the motor command that will achieve this desired sensory outcome. But because you don’t know which exact muscle activations are needed, the brain learns this mapping using the feedback-error learning model. The difference between the *desired state* (where you wanted the ball to go) with the *estimated state* (where it actually went) is fed to the feedback controller, which turns the sensory error into a motor error. This feedback motor command reflects how much correction was needed. That correction signal is used to train the inverse model. The motor command is the sum of the motor error and the feedforward motor comment (output from the inverse model).

Through repeated swings, the inverse model learns to generate the correct motor commands on its own, allowing you to reliably hit the ball where you intend.

Examples of “theoretical” exam questions (from last year’s exam)

Social cognition

- Describe how Theory of Mind abilities can be experimentally tested in 6-month old babies. Explain a task that could be used, and what it measures.
- Answer:

Theory of Mind in 6-month-old infants can be tested using anticipatory looking using an implicit false-belief task, since infants do yet have language to perform an explicit false-belief task like the Sally-Anne task. Infants watch an agent place a toy in one of two boxes. While the agent cannot see, the toy is moved by a puppet. When the agent returns, infants’ anticipatory eye movements reveal where they expect the agent to search. Infants reliably look toward the location consistent with the agent’s false belief, showing they can represent another’s belief and use it to predict actions. The task measures belief attribution using looking behavior/eye tracking rather than explicit responses.

Examples of “theoretical” exam questions

Social cognition

- How can social interaction be “more than the sum” of individual contributions?
- Answer:

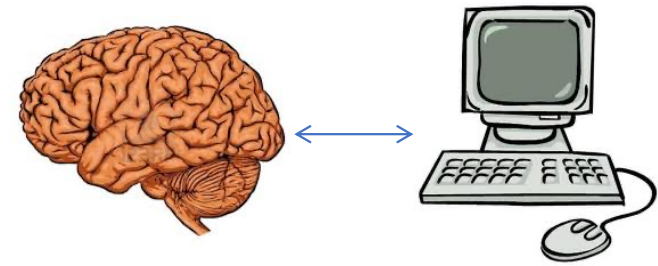
Social interaction can become an autonomous process with its own dynamics, not reducible to the behaviours of each individual. When two people coordinate their actions over time, this coordination shapes both how the interaction unfolds and how each person continues to act. For example, in the charades scenario, the meaning of the gesture (“window”) is jointly constructed through back-and-forth adjustments, misunderstandings, and refinements. The interaction itself guides both partners, creating new shared meaning that neither individual could generate alone. Thus, the interaction becomes more than just the sum of two isolated behaviors.

Embodied cognition learning objectives

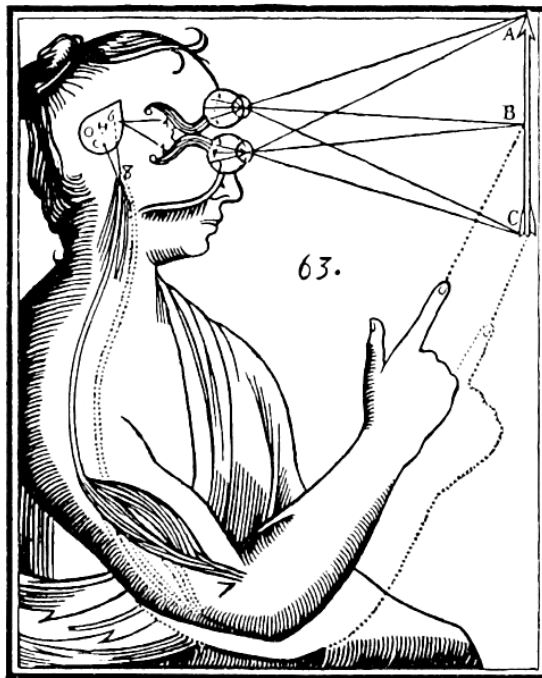
- You should be able to:
 - explain the traditional views of cognition, and why these were challenged
 - describe what embodied cognition is, and how this is relevant to AI, including examples from robotics
 - describe the 6 claims of embodied cognition, and be able to give an example of each one
 - discuss problems with some of the claims of embodied cognition
 - describe what psychophysiology is and why embodied cognition underlies this field
 - describe what heart rate variability is, and how it is measured and interpreted
 - analyze HRV using frequency domain methods, and explain how it relates to human cognition
 - analyze synchrony between people's physiological signals, and interpret results in the context of human cognition

Traditional views of cognition

- cognition as a computational process restricted to the brain
- brain is an abstract information processor, with internal representational states, whose connections to the outside world are of little theoretical importance
- the brain is at the seat of cognition
- perception & action not relevant to understanding “central” cognitive processes, but peripheral input/output devices
- experiments designed to reveal the content of representational states or uncover the algorithms that map input onto output



Mind-body dualism: the sharp contrast to embodied cognition



- mind-body dualism: mind and body are distinct and separable
- mental phenomena are non-physical

“there is a great difference between mind and body, inasmuch as body is by nature always divisible, and the mind is entirely indivisible... the mind or soul of man is entirely different from the body.”

- Rene Descartes
(1600s)



Challenging this traditional approach

“The body is the vehicle of being in the world, and having a body is, for a living creature, to be involved in a definite environment, to identify oneself with certain projects and be continually committed to them”

- Maurice Merleau-Ponty (1962)



Challenging this traditional approach

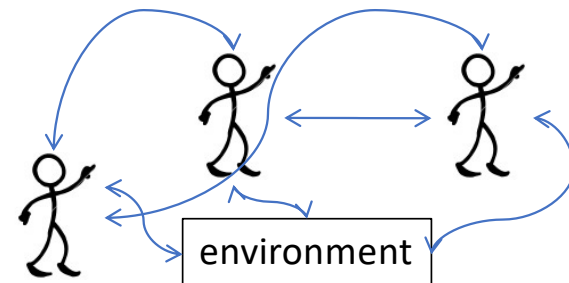
“Biological brains are first and foremost the control systems for biological bodies. Biological bodies move and act in rich real-world surroundings”

- Andy Clark (1998)



Challenging this traditional approach

- Cognitive processes and their context unfold continuously and simultaneously in real-time
- In continuous interaction with the environment
- Cognitive system includes nervous system, body, environment, and all possible interactions between them



What is embodied cognition?

- the body (and its interactions with the environment) is deeply integrated into cognition, and has a central role in shaping the mind: how we think, learn, perceive, speak, remember, problem solve, interact
- most of our thinking is embodied: if we move differently, we end up thinking differently
- mental processes are not ONLY computational processes

AI fails when there is a need for embodied cognition

Task:
Choose which piece to
move where

high-level task



Task:
Physically manipulate
each piece and move it
around the board

low-level task



**Problems that are difficult
to solve for AI are
associated with the
interface to the real world**

*“The body, in a very simple
way, is the foundation for
intelligent and cautious
action”*

– Joshua Bongard, NYT
(April 11th, 2023)

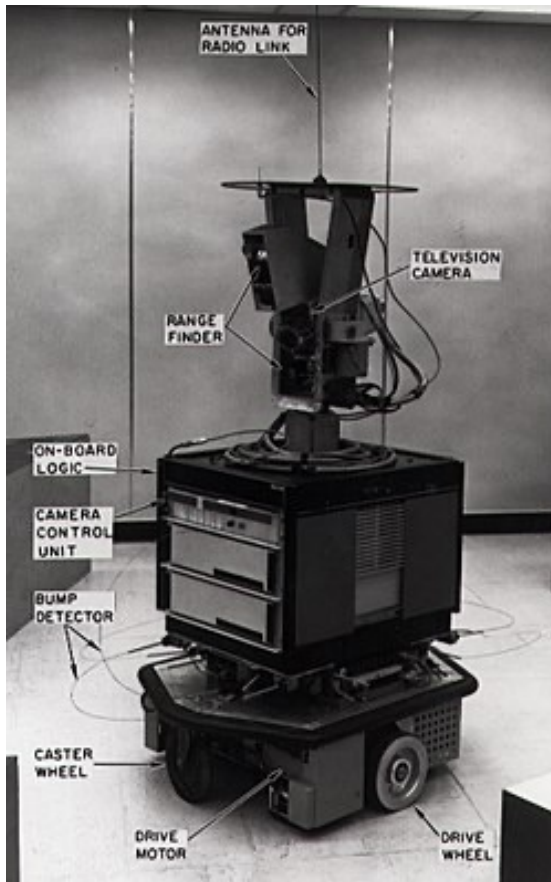
*Author of “How the body
shapes the way we think: A
new view of intelligence”*



Artificial intelligence as representation

- AI traditionally founded on the issue of representation:
 - computer models of abstract symbol processing
 - independent information processing units with internally stored rules
 - specialized subproblems that represent knowledge, natural language understanding, vision, etc. rather than attempting to replicate a bigger scope of human intelligence
 - no meaningful interface to the environment
- Can human intelligence be decomposed into such subpieces?
- Most ML systems even today learn by passively absorbing large data sets instead of learning through interaction with the world

First mobile robotic AI (1966-1972)



- programmed to navigate through a room, avoid blocks of various shapes and colours
- cycled through iterations of sense-model-plan-act sequences
- able to reason about its actions
- combined computer vision, NLP
- is this an AI that computes symbolic representations or is embodied in the world, interacting with its environment?
- computes representations:
 - camera senses the environment, while a computer builds a symbolic model of it from the camera images
 - program combines the symbolic descriptions to create plans for a course of action
 - no bidirectional interaction with the environment?

Rodney Brooks' robots: AI without representation

- do not need to “represent” the world: no representation should stand between the sensory signals and the robot’s behaviour
- connect perception to action directly
- evolve through continuous loops: the body moves, which changes the stimulation its sensors receive, which directly causes new movement, etc.
- Robot Allen: sensors connect directly to 3 behaviour-generators - Avoid, Wander, Explore
 - Avoid causes robot to stop, turn, and proceed, when detecting objects in its path
 - Wander sends robot along a random heading
 - Explore steers robot toward a selected target
 - 3 “activities” compete with each other

BigDog robot (Boston Dynamics)



- BigDog can solve complex physical adaptive problems, responding to the environment
- Very little “brain”
 - built with sensors and actuators
 - nothing dictating the form of motion
 - in constant interaction with the environment
 - dynamically stable
 - wide range of locomotion behaviours

Claims of embodied cognition:

1. cognition is situated

- cognitive activity takes place in the context of a real-world environment, and it inherently involves perception and action
- involves interaction with the things that the cognitive activity is about
- takes place in the context of task-relevant inputs and outputs
 - perceptual info continuously comes in and affects the cognitive processing
 - the executed motor activity affects the environment
 - e.g., driving, holding a conversation, spatial cognition etc.
- problem with this: planning, remembering, day-dreaming, imagining, all take place decoupled from immediate interaction with environment (all “offline” cognitive activities)

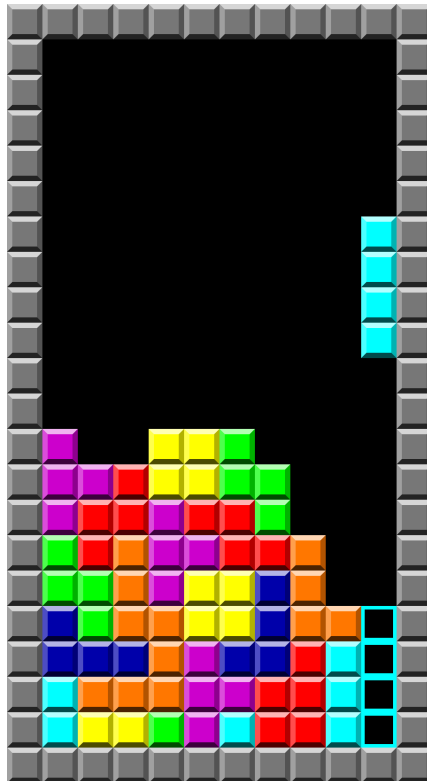
Claims of embodied cognition:

2. cognition is time pressured

- situated agents must deal with the constraints of “real time”
- traditional AI models can build up and manipulate internal representations at their own pace, but real agents often must cope with the environment with time pressure
- in AI, greater cognitive flexibility can be built up with procedures built for real-time interactions with the environment
- e.g., remember real-time coordination examples from social cognition, such as tapping or music – people don’t need to build a full mental model of other people or the environment; they predict and adapt quickly
- but, human cognition often does not cope well with time pressure: remember forward models in motor control – used because sensory feedback is delayed, and there is no time to build a full model of the environment and its states

Claims of embodied cognition:

3. we offload cognitive work onto the environment



- e.g., Tetris: the players use actual rotation and translation movements to simplify the problem to be solved, rather than mentally computing a solution and then executing it
- Rather than mentally store and manipulate all the situational details, we physically store and manipulate those details out in the world
- Other examples?

Claims of embodied cognition:

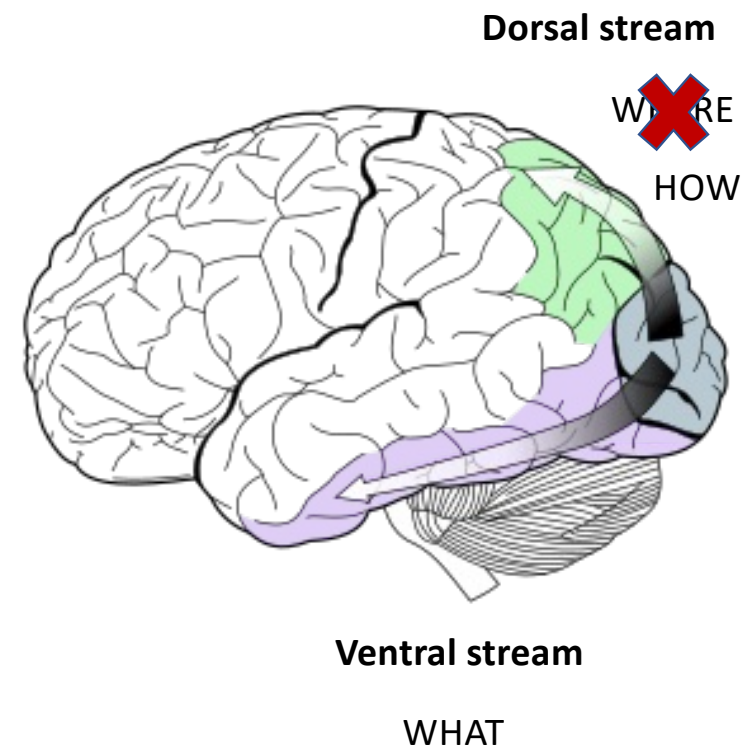
4. the environment is part of the cognitive system

- cognition is not an activity of the mind alone, but distributed across the mind, body, and environment
- to understand cognition, we must study the situation and agent together as a single, unified unit: the mind alone is not a meaningful unit of analysis
- quite radical – how do we define the boundaries of this “unit” or system?
 - the constitutive elements of the unit/system would change every time we move to a new location or begin interacting with a different set of objects or people
 - how do we find underlying principles and regularities of cognition and cognitive mechanisms?

Claims of embodied cognition:

5. cognition is for action

- recall the lecture on motor control: we have a brain for the sole purpose of producing flexible and adaptive movements
 - vision, memory, other cognitive processes evolved in the service of perception and action – for the purpose of driving or suppressing future movements
- e.g., visual perception
 - traditional view: vision builds up an internal representation of the perceived world, sends it to “higher” cognitive areas to decide what to do with this
 - 2 distinct pathways: ventral & dorsal
 - Ventral: “what” route, recognizing objects
 - Dorsal: “where” route, locating objects in space
 - recent views: Dorsal route is the “**how**”, for acting on objects, linking vision to action; function: to serve visually guided actions (reaching/grasping)
 - neurons in dorsal stream correspond not only to presence of objects, but whether objects are relevant for a particular task
 - prioritizes visual information, relevant to action & goals



Claims of embodied cognition: 6. offline cognition is body based

- we manipulate the environment to help us think about a problem, but this need not rely on overt actions
 - e.g., counting fingers, using the motor programs as mental imagery with no overt movement to assist counting
- many abstract cognitive activities make use of sensorimotor functions in this covert way, running offline, decoupled from physical inputs/outputs, to assist thinking and knowing:
 - Mental imagery
 - Working memory
 - Reasoning and problem solving
 - offloading information onto perceptual and motor control systems in the brain, rather than the environment
- rather than the mind operating to serve the body, the body serves the mind

Examples of embodied cognition from social cognition lectures?

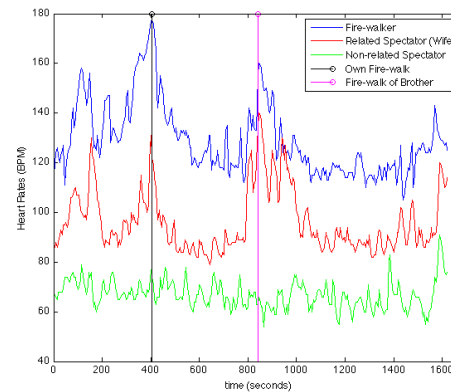
- joint action studies – action coordination is cognition; influence of other people's bodily movements on one's own movements and cognition
- emotional contagion via facial mimicry
- mirroring systems (i.e. mirror neurons), simulation theory: action understanding and social cognition begins at this primitive level of motor planning

Psychophysiology as embodiment

- psychophysiology is the study of the relationship between physiological signals recorded from the body to mental processes and disorders
- cognitive and psychological processes have a physiological basis (cognition is not just something that happens in the brain)

Psychophysiology as embodiment

- e.g., mental stress can present itself as bodily pain, chest pain, digestive problems, ect.
- e.g., Social relationships are embodied:
 - Physiological (heart-rate) synchronization between individuals linked to social bonds, group cohesion
 - HR synchrony found between parents and infants, romantic couples, during rituals between socially affiliated performers and spectators

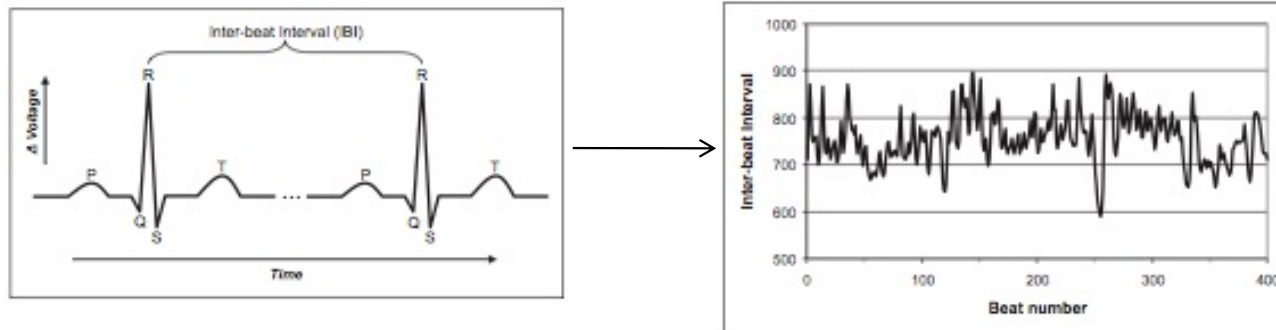


Established physiological “correlates” to psychological states

- Electrodermal Response (EDA/GSR)
 - arousal, excitement, motivational activation
- Electrocardiography (ECG)
 - Heart rate (HR): attention, cognitive resource allocation, engagement
 - Heart rate variability (HRV): emotional arousal, emotional regulation, attention
- Facial electromyography (EMG)
 - emotional valence

Heart rate variability (HRV)

- A healthy heart is not a metronome
- Change in heart period in consecutive beats (using instantaneous HR or R-R intervals)
- Influenced by respiration (HR increases when inhaling, decreases when exhaling to optimize oxygen uptake): respiratory sinus arrhythmia (RSA)



HRV: Sympathetic vs. Parasympathetic

Sympathetic

- ↑ heart rate
- dilates pupil
- Inhibits salivation
- Relaxes bronchi
- Inhibits digestive activity
- Relaxes bladder

Parasympathetic

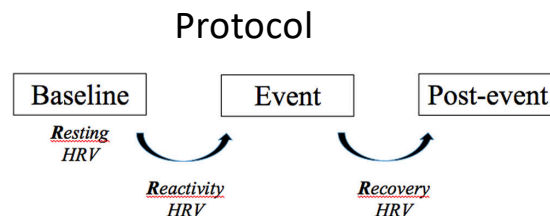
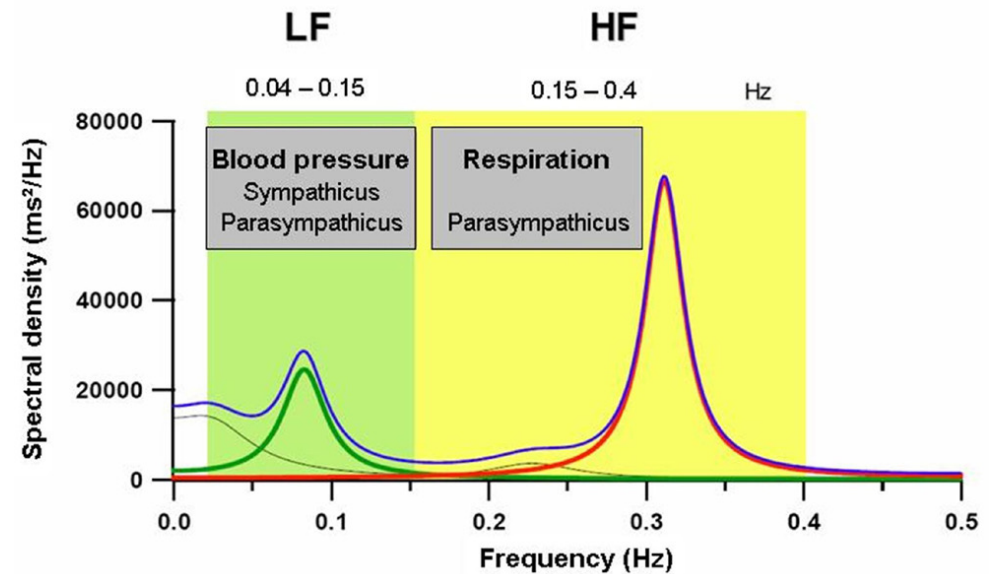
- ↓ heart rate (via vagus nerve)
- Constricts pupil
- Stimulates salivation
- Constricts bronchi
- Stimulates digestive activity
- Contracts bladder

- Work in a complementary manner to achieve homeostasis, with the heart receiving both signals in a push-pull manner

Heart rate variability (HRV)

Frequency-domain methods

- Power spectral density (PSD)
- Low frequency (0.04-0.15 Hz) -> sympathetic & parasymp.
- High frequency (0.15-0.4 Hz) -> parasympathetic
- LF/HF -> sympathovagal balance?

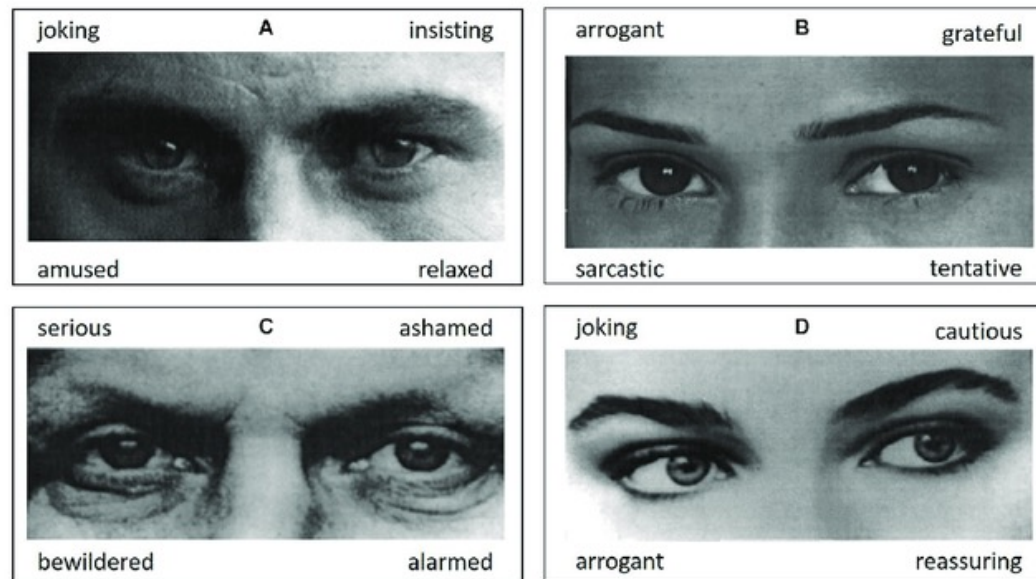


HRV and cognition

- emotional arousal
 - HF-HRV decreases under conditions of time pressure and emotional strain, elevated anxiety state, etc.
 - HRV lower in individuals who worry more, and in anxiety disorders, PTSD, etc.
- emotion regulation
 - people with higher HRV at rest can provide more appropriate emotional responses, and better regulate negative emotions
- attention
 - HF-HRV associated with attentional control: high anxiety individuals (with low HF-HRV) have poorer attentional control
 - high HF-HRV associated with better performance in learning tasks

HRV and emotion recognition

- Polyvagal theory: ANS evolved to support the survival, reproduction, and social engagement of the species; hence HRV may provide a sensitive marker of one's ability to respond to and recognize social cues



- resting HF-HRV positively correlated with performance on the “Reading the Mind in the Eyes” test
- you can try the test here: <http://socialintelligence.labinthewild.org/mite/>

Class experiment from week 1 – HRV (part I)

- You participated in 3 different games in teams with your classmates: people bingo, deserted island, and tower game
 - Deserted island and tower games were both team based
 - The class was split into 2 sessions (A & B, session B only did people bingo and tower game)
 - N = 91 students total
- Everyone also filled out a social network questionnaire, checking off their close friends and friends in the class
- Data we recorded:
 - Heart rates from everyone (and accelerometry, but this will not be used), also during baseline
 - Which participants were team members
 - Which participants are friends, which are not

Class experiment from week 1 – HRV (part I)

- Is HRV modulated by the game/task (in relation to baseline)?
- What does it mean if we see modulation in HRV (in the context of physical & cognitive processes)?

Class experiment hypotheses

- Group activities/interactions decrease parasympathetic activity (HF-HRV)?
 - lower HF-HRV for each game in relation to baseline
- Interpretations:
 - while physical stress will increase sympathetic activity, mental stress decreases parasympathetic activity – stress (physical or mental) will result in a higher sympathetic-parasympathetic balance of our ANS
 - one potential mechanism may be through deregulation of breathing
 - mental stress, cognitive load, attention are embodied
 - But: there is a lot of movement and physical activity during the games – this will certainly affect HRV.

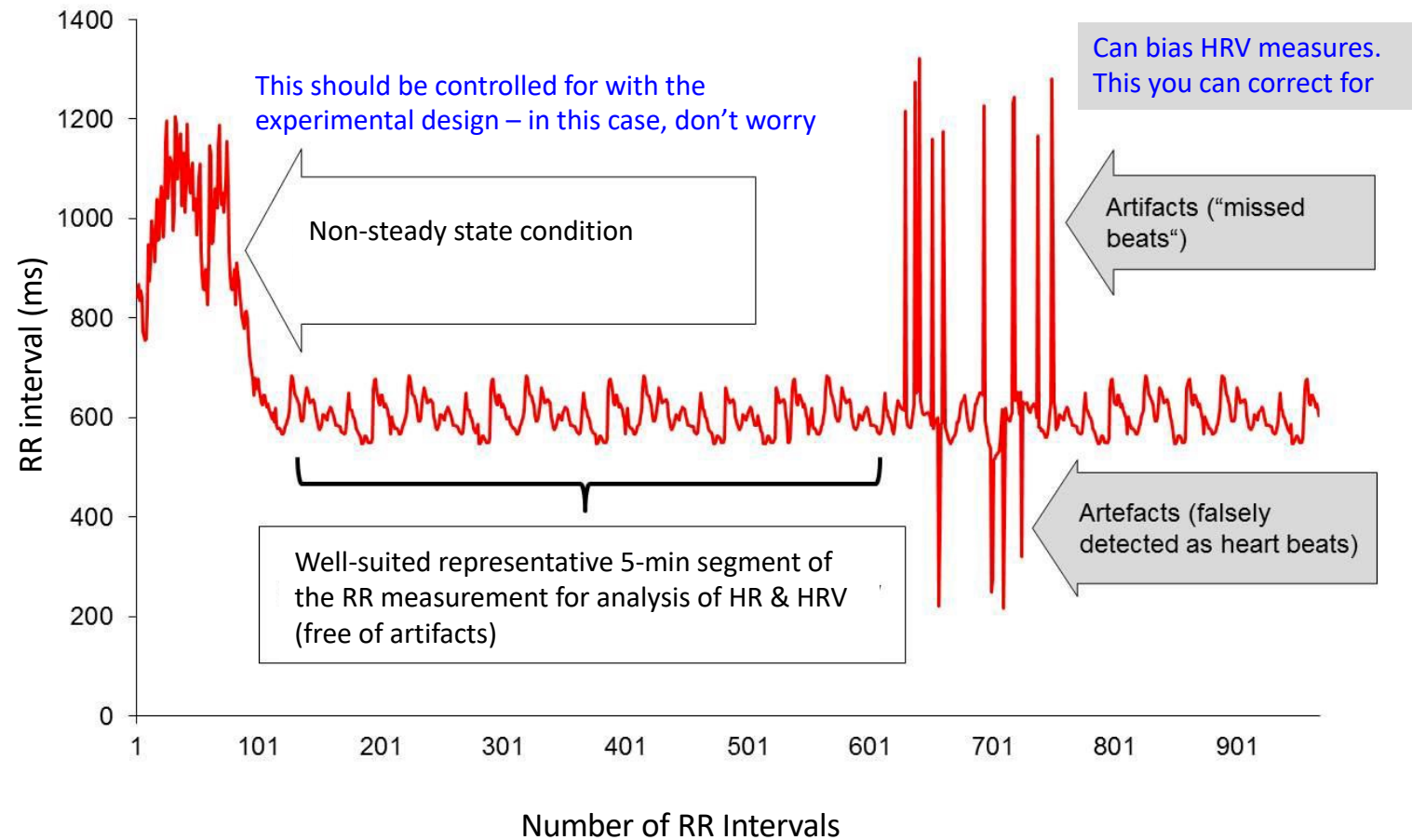
HRV analysis

step 1: find the relevant data segments

- you have the start time of recording (in HH:MM:SS): convert these to seconds
- you should also have time stamps for each game (they are available in the notebook):
using R-R intervals and start time of recording, work out at which R-R your task conditions begin and end
- isolate your data segments, e.g., baseline, game 1, game 2, game 3

HRV analysis

step 2: inspect your data



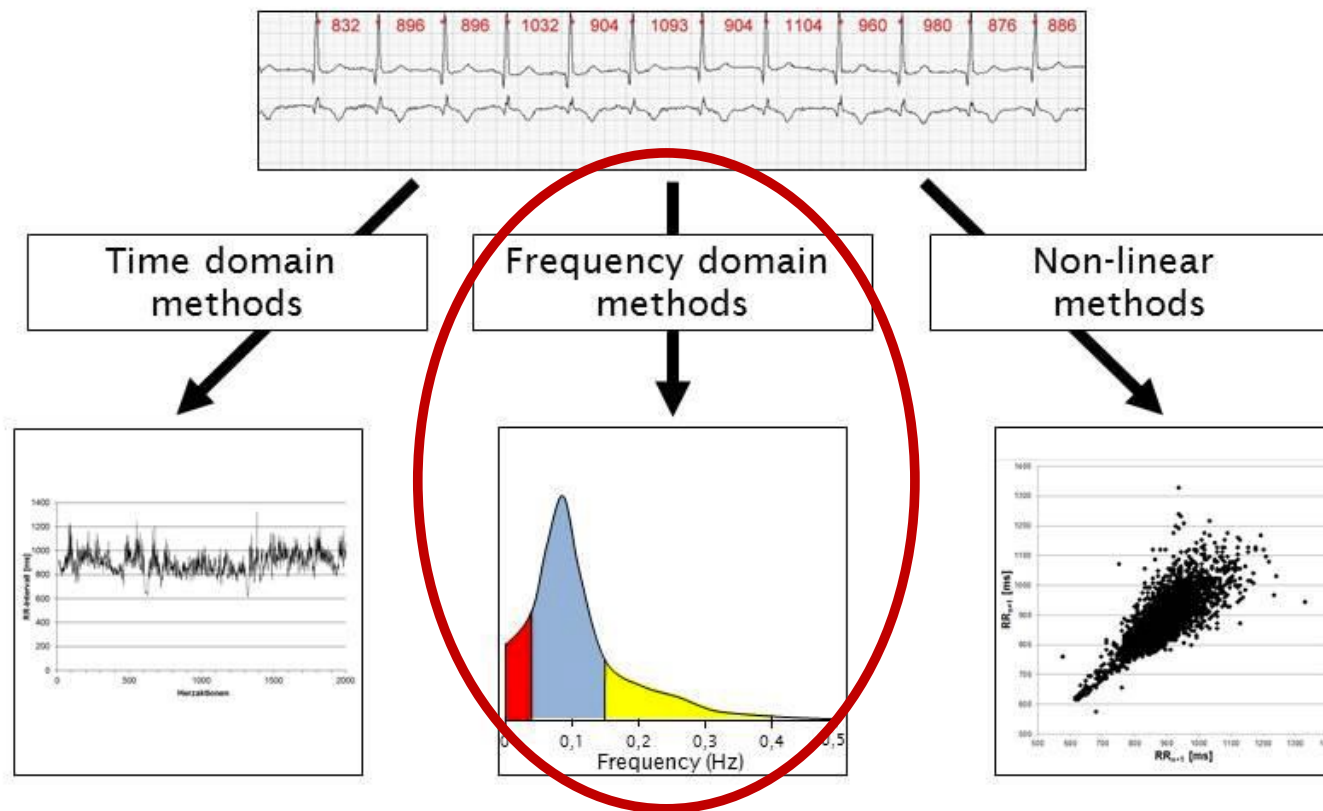
HRV analysis

step 3: correct for missed/false detected beats

- **Note: you can skip this step for the sake of this class, as it can be quite time consuming (just know that it is generally necessary for spectral HRV analyses)**
- Falsely detected beats (e.g., $R-R < 0.3$ seconds) that are sharp peaks: delete
- Missed beats:
 - linear interpolation: fit straight line over abnormal R-R intervals to obtain new values (i.e., from 3 preceding intervals)
 - or cubic spline interpolation (3rd degree polynomial)

Calculating HRV

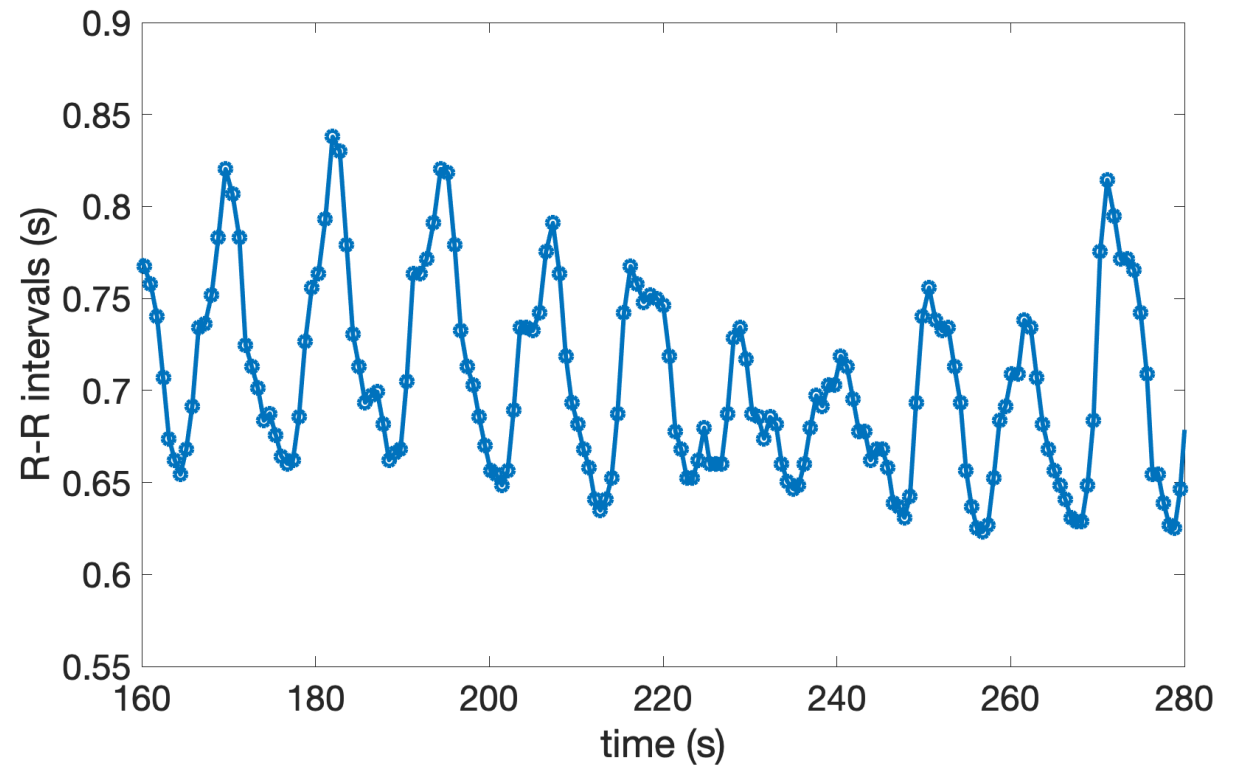
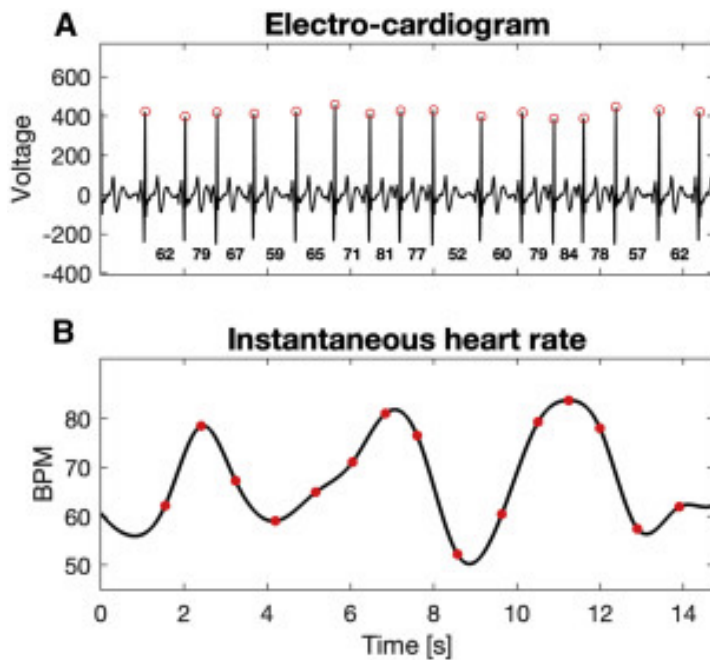
Heart rate variability (HRV)



HRV analysis

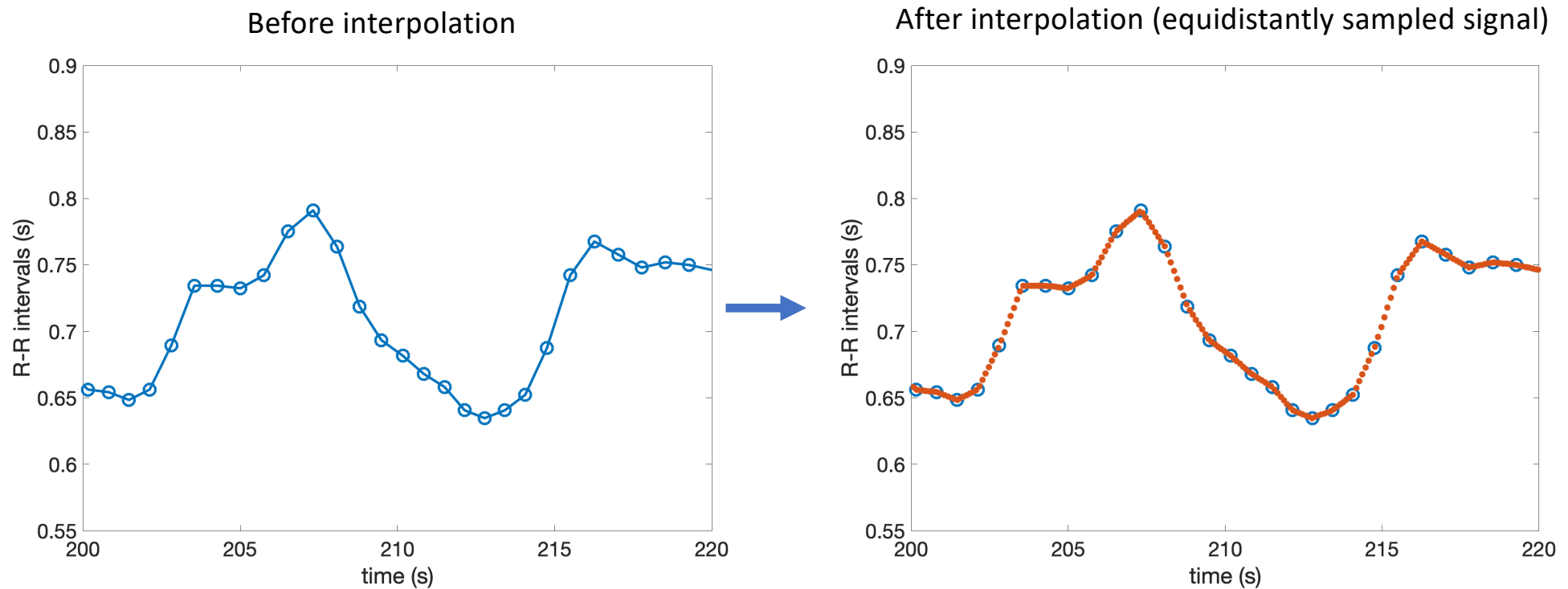
step 4: interpolate/resample R-R intervals so they are sampled in time

Non-interpolated R-R intervals (time step between beats is not constant)



HRV analysis

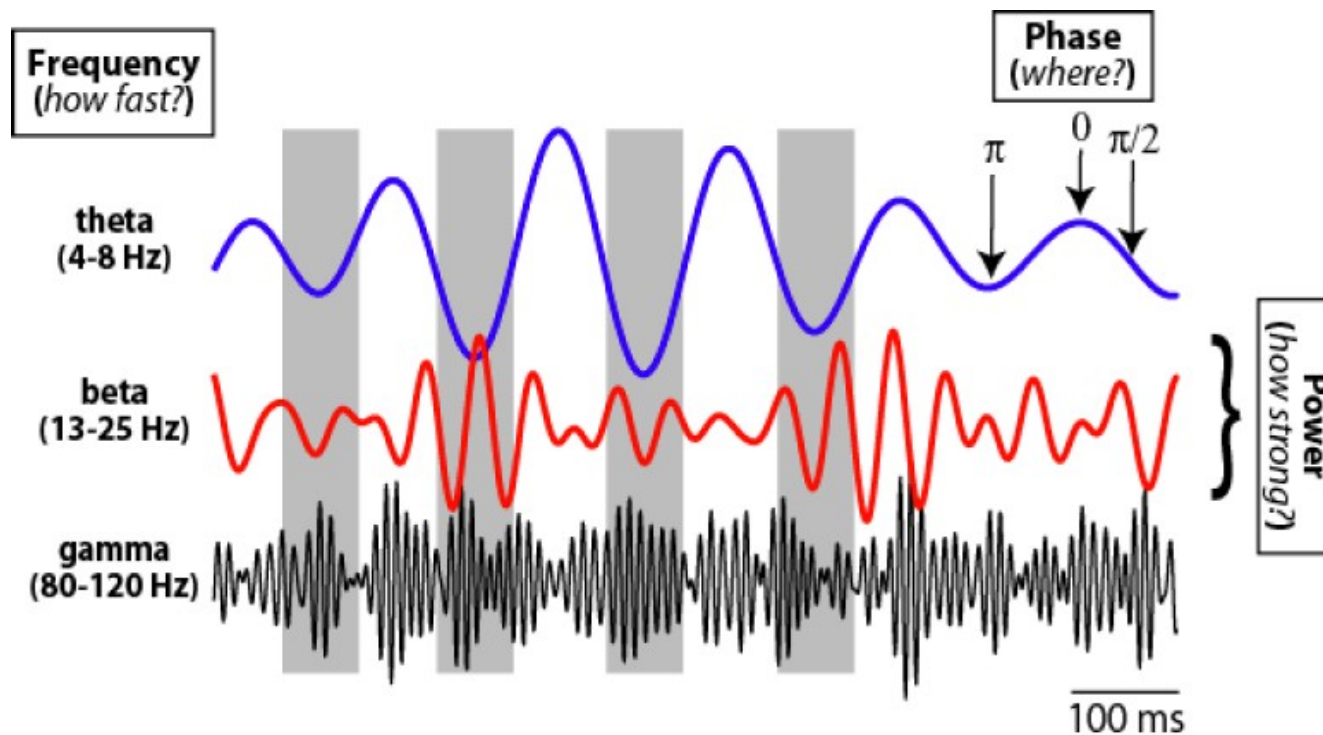
step 4: interpolate/resample R-R intervals so they are sampled in time



Choose a sampling frequency of 10 Hz, and linear interpolation
Use `interp1d` (Python) or `interp1` (MATLAB)

HRV analysis

step 5: frequency domain analysis

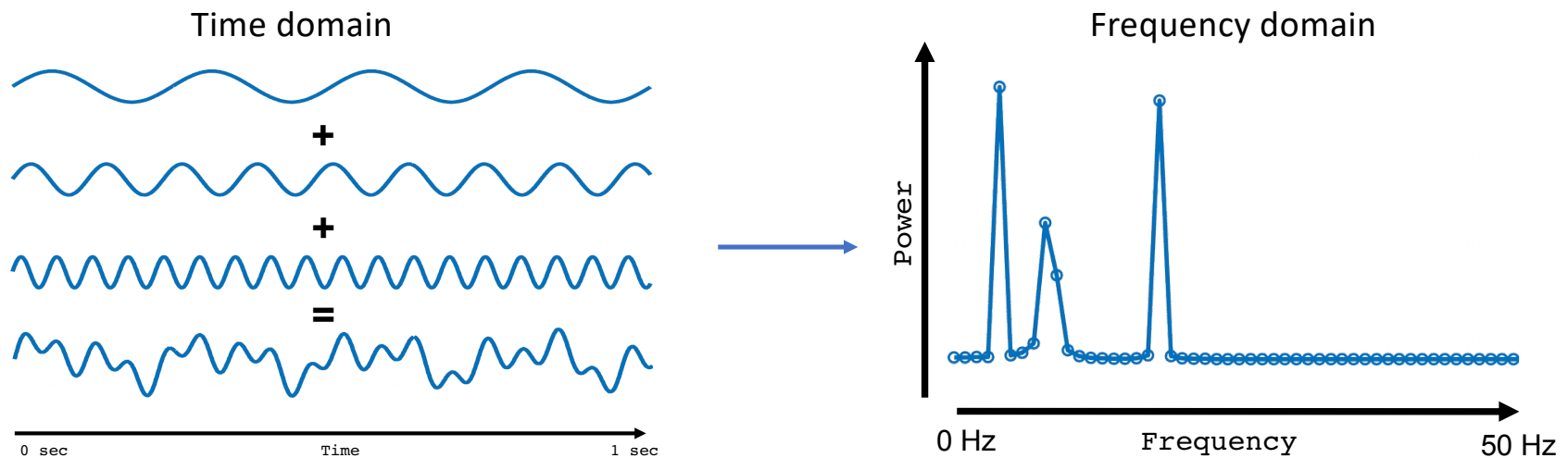


Time Domain -> Frequency Domain?

HRV analysis

step 5: frequency domain analysis

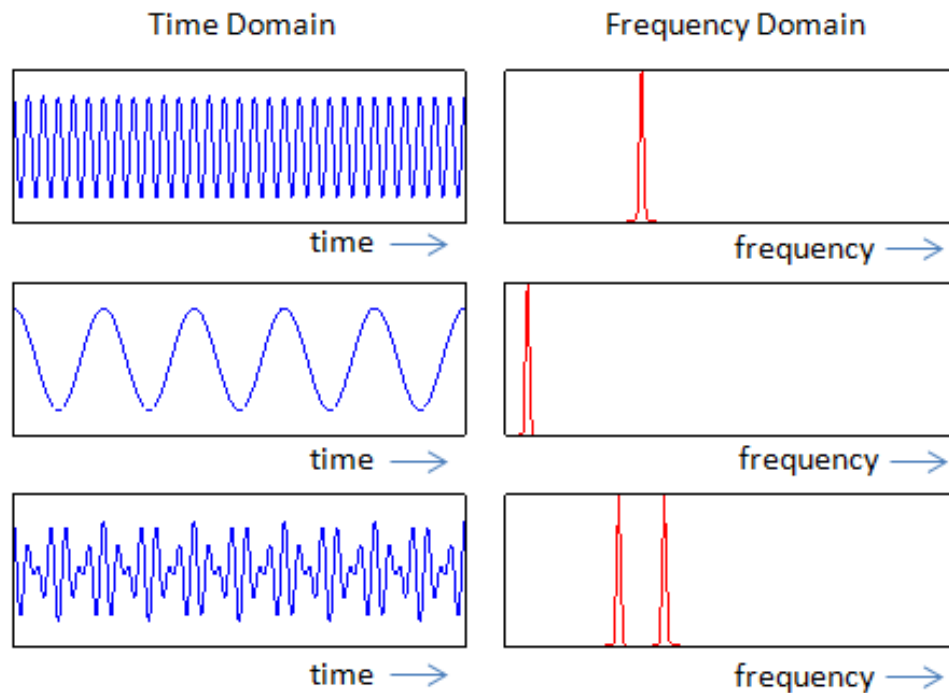
- (fast) Fourier Transform:
 - mathematical transformation employed to transform signals from time domain to frequency domain
 - breaks a signal into an alternate representation, characterized by sines and cosines (any periodic signal can be rewritten as the weighted sum of sinusoidal functions)



HRV analysis

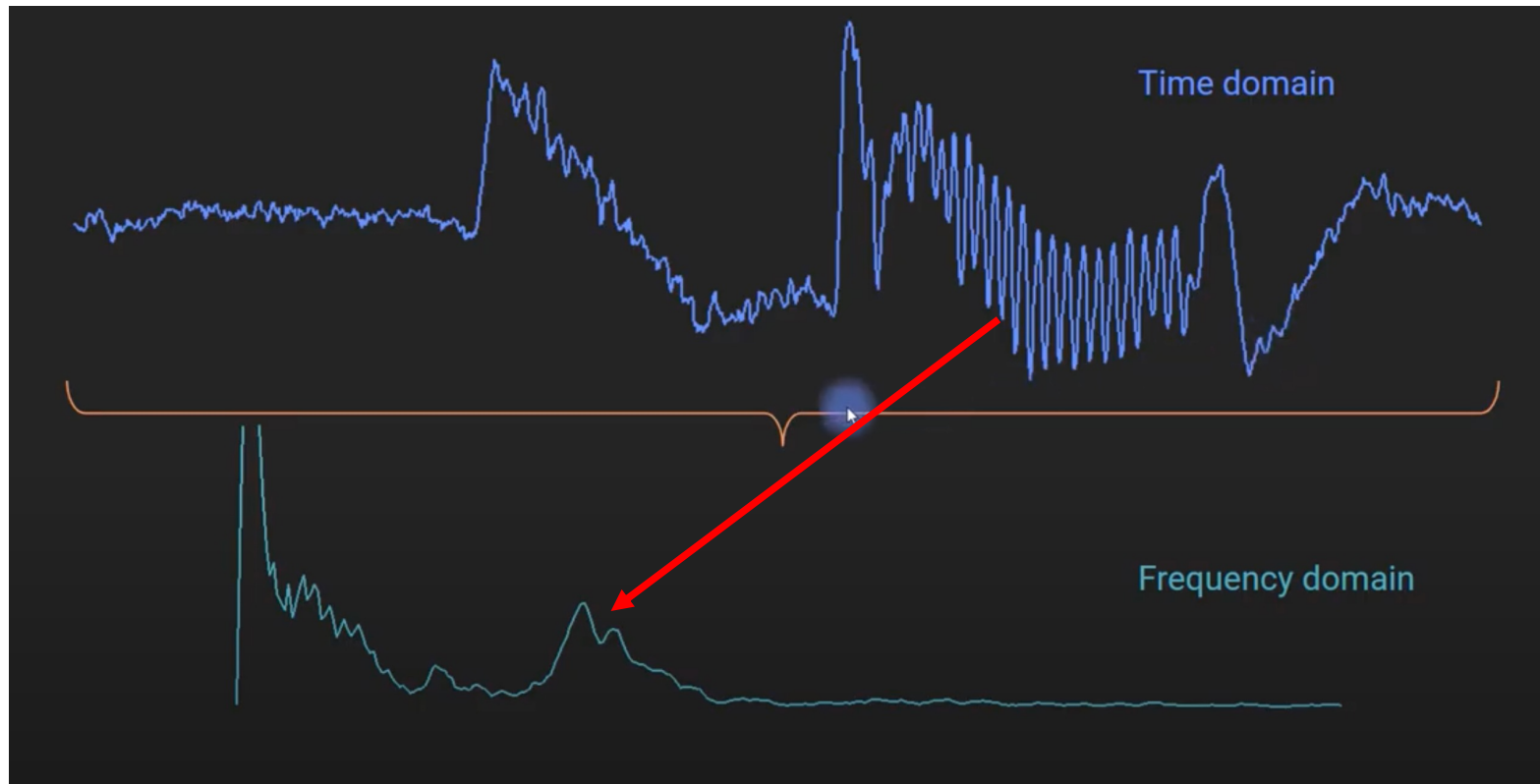
step 5: frequency domain analysis

estimate power spectral density (PSD)



- peaks in the frequency domain correspond to an oscillatory component in the time domain
- power spectral density: how much of the signal's power is concentrated in its different frequency components
- the spectral peaks in HRV power spectrum reflect the amplitude of the heart rate fluctuations present at different oscillation frequencies
- e.g., deep breathing will induce an oscillation in the heart rhythm, and peak in the power spectrum will appear at the breathing frequency

FFT method

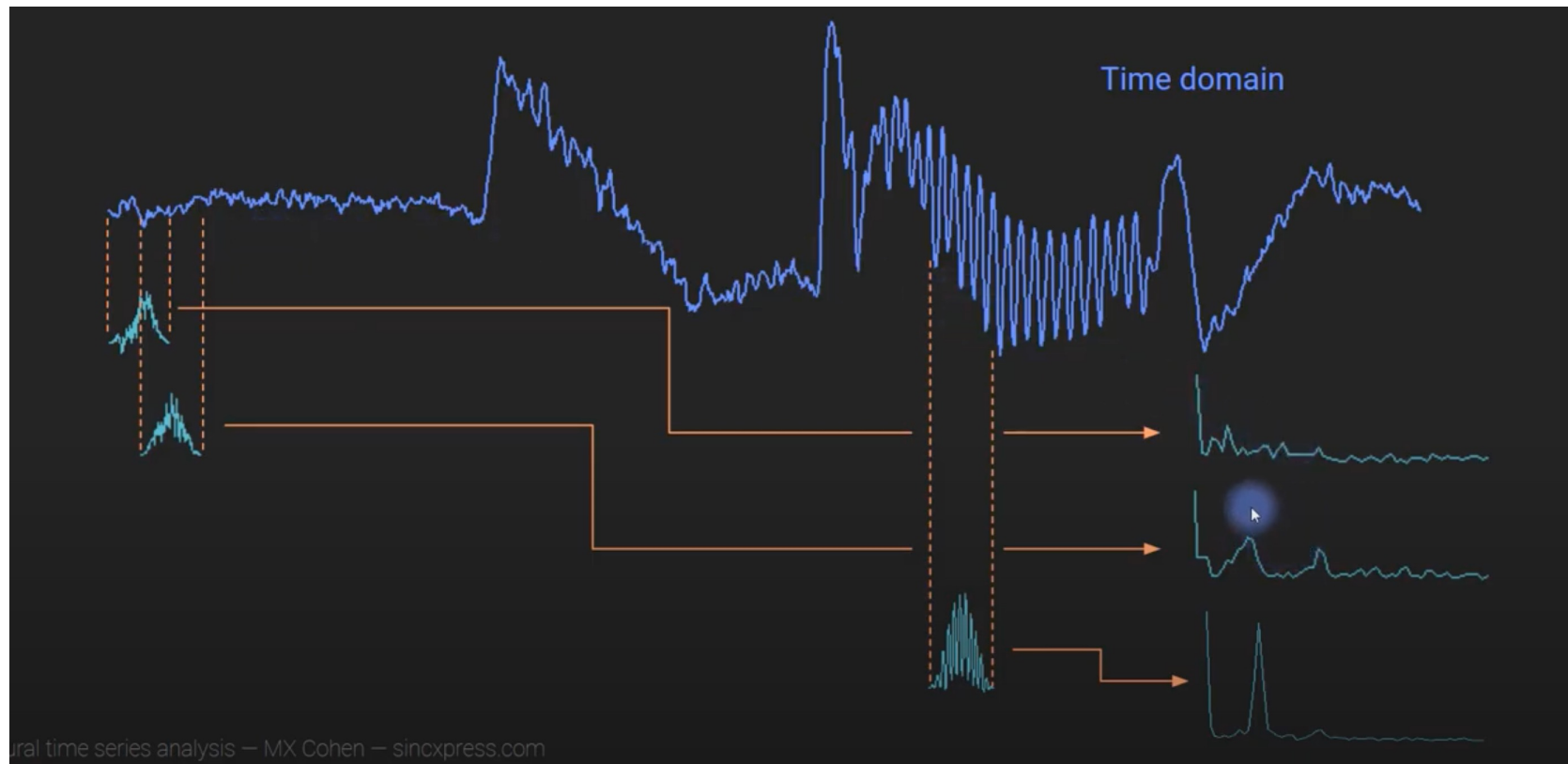


More sensitive to noise and non-stationarities

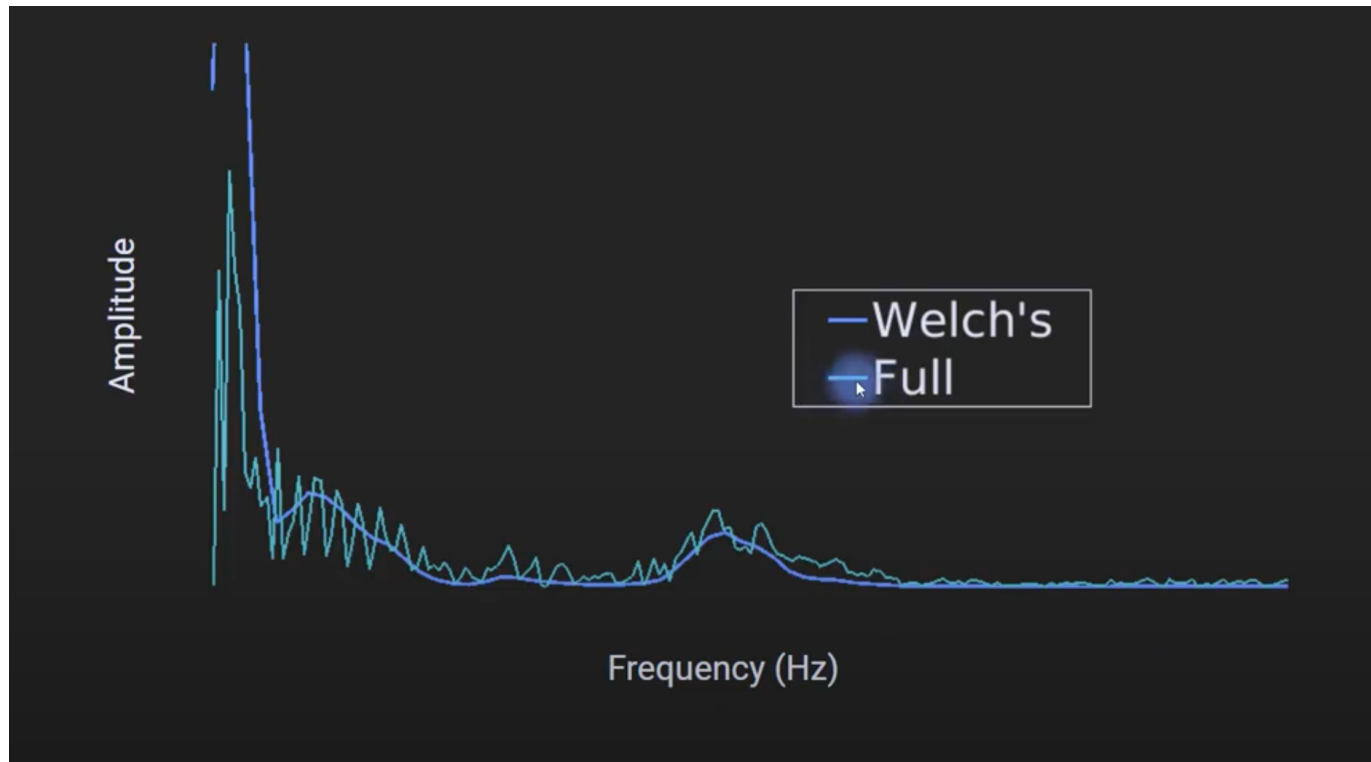
HRV analysis
step 5: frequency domain analysis
estimate power spectral density (PSD)

- (fast) Fourier Transform:
 - mathematical transformation employed to transform signals from time domain to frequency domain
 - breaks a signal into an alternate representation, characterized by sines and cosines (any periodic signal can be rewritten as the weighted sum of sinusoidal functions)
 - spectral estimate of the entire signal – more sensitive to noise, less able to cope with non-stationarities in the signal
- Welch's periodogram
 - uses localized averaging to reduce noise
 - gives a smooth spectral decomposition, and hence easier to interpret compared to FFT

Welch's method



FFT vs. Welch's method



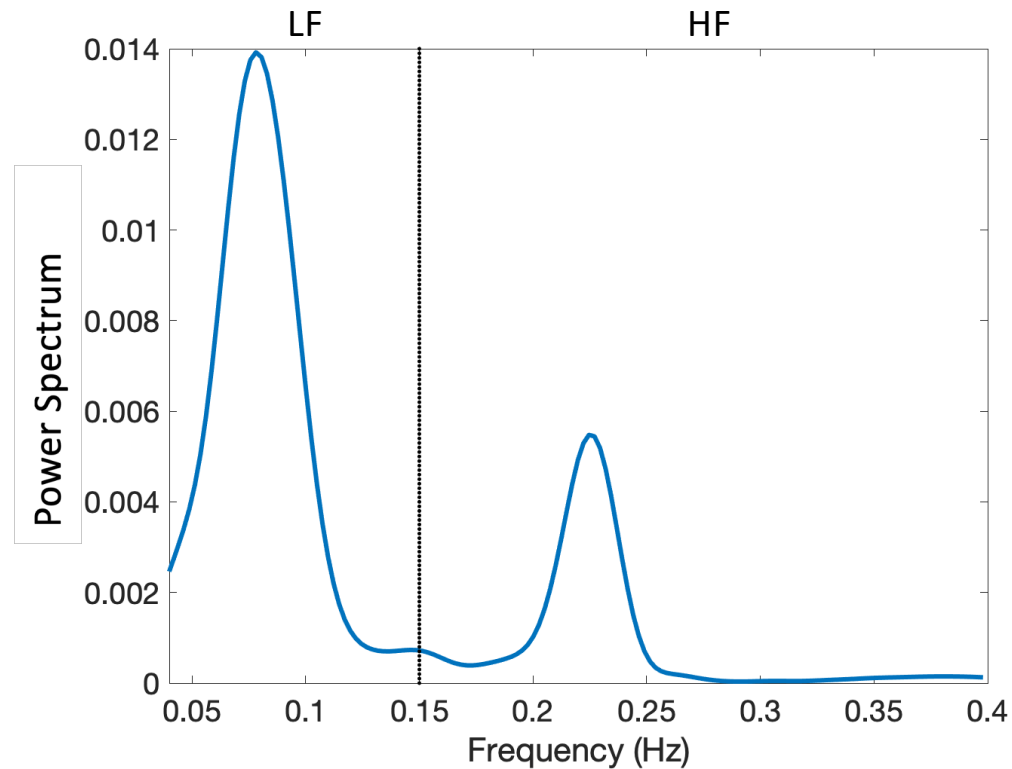
Welch's method gives a smoother power spectrum

HRV analysis
step 5: frequency domain analysis
estimate power spectral density (PSD)

- Use Welch's periodogram (welch in Python, pwelch in MATLAB)
- You will use a Hamming window for tapering, to minimize edge effects (window length and overlap defined in the exercise)
- For more details of the method, see: <https://www.youtube.com/watch?v=YK1F0-3VvQI>

HRV analysis

step 6: plot the power spectrum (for each condition)
and integrate in the HF range to calculate HF-HRV



To estimate the PSD in
HF (i.e., integrate), use
`np.trapz` (Python)

HRV analysis

step 7: simple statistical comparisons

- Plot your data (HF-HRV across each game or baseline) using e.g., box plots or raincloud plots
- Do simple t-tests (or rather, non-parametric tests as data won't be normally distributed) to compare conditions, e.g., each game to the baseline
- Discuss:
 - how do you interpret more HF-HRV in one condition relative to another?
 - how is this related to cognition (cognitive processes)?
 - how does it relate to embodied cognition?